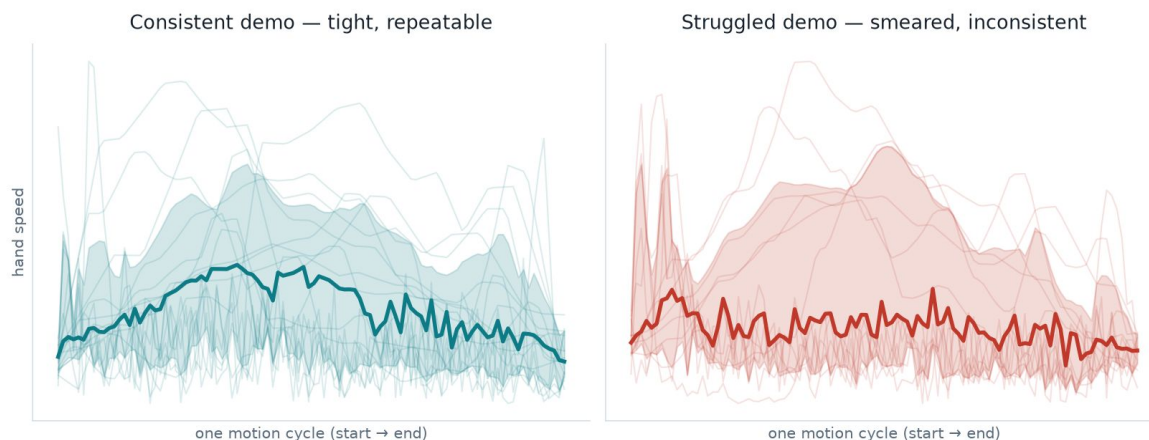


Verifying human demos with signal-integrity methods

Motivation: Robots learn by imitation, so a single fumbled demo, but that looks subtly fine on video, quietly teaches the wrong thing. At dataset scale, no one can watch every clip; bad demonstrations have to be found automatically. We find them from hand motion alone: no AI judge, no training.

OUR KEY APPROACH: THE EYE DIAGRAM

Inspired by signal-integrity engineering, we overlay every repetition of a motion on top of itself. A consistent action stacks into a tight band that appears as an eye; a struggle does otherwise. With this data, we turn it into a quality score.



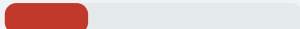
QUALITY-AWARE SEARCH

Ask in plain language and get clean, quality-ranked demos back. The failure score re-ranks so bad clips sink.

🔍 find successful washing dishes demo

wash_dishes  clean

wash_dishes  clean

wash_dishes (fumble)  sinks

Overview: Deterministic · No training · No LLM · Scored 5,000 demos on Modal in ~1 min, 0 errors · Adaptive, validated on real video (precision@10).